

SCS1206 / ENH1301 Laboratory I

Comprehensive Past Paper Synthesis & Short Notes

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Contents

1	Computer System Architecture and Fundamentals	3
1.1	Data vs. Information	3
1.2	Characteristics of Computers	3
1.3	Generations of Computers	3
1.4	Storage Units and Calculations	3
1.5	System Layers and Operating System Kernel	4
2	GNU/Linux Architecture and Directory Hierarchy	4
2.1	Filesystem Hierarchy Standard (FHS)	4
2.2	Path Navigation	4
2.3	Symbolic and Hard Links	5
3	Linux File Permissions, Ownership, and Security	5
3.1	Understanding <code>ls -l</code> Permission Output	5
3.2	Changing Permissions with <code>chmod</code>	5
3.3	File Ownership Commands	6
3.4	Essential Security and Credential Files	6
3.5	Procedure to Recover Forgotten Root Password	6
4	Disk Management, Partitioning, and File Systems	6
4.1	Master Boot Record (MBR) Partitioning Types	6
4.2	Disk Device Naming Scheme	7
4.3	Disk Management Commands	7
5	Process Management and System Monitoring	7
5.1	System Monitoring Tools	7
5.2	Understanding Load Average Values	7
5.3	Job Control Key Combinations and Commands	8
6	Package Management and System Maintenance	8
6.1	Debian/Ubuntu Package Management (APT)	8
6.2	RedHat Package Management (RPM)	8
6.3	Methods of Installing Software on Linux	8
7	Networking and System Information Commands	9
7.1	Networking Commands	9
7.2	System Hardware Information	9

8	BASH Shell Features, Commands, and Pipelines	9
8.1	Common File Manipulation Commands	9
8.2	BASH Shortcuts and Environment Operators	10
8.3	Advanced Text Processing Pipeline Commands	10
9	BASH Scripting and Automation	11
9.1	Script Positional Parameters	11
9.2	Required BASH Script Solutions	11
9.2.1	Script 1: User Greeting (hello)	11
9.2.2	Script 2: Word Count Display (whatsin)	11
9.2.3	Script 3: Arithmetic Addition (add)	11
9.2.4	Script 4: File Existence and Permission Checker (files)	11
9.2.5	Script 5: Argument Counter and Listing (args)	11
10	Vim Text Editor Syntax and Operations	12
10.1	Vim Modes	12
10.2	Key Combinations in Normal/Command Mode	12
10.3	Window Splitting and File Navigation	12
10.4	Search and Replace Syntax	12
10.5	Executing Shell Commands and Key Mapping	13
10.6	Spellchecker Navigation	13
11	Git Version Control System	13
11.1	Advantages of Source Code Version Control	13
11.2	The Three Areas Defined in Git	13
11.3	Essential Git Commands	13
12	LaTeX Technical Writing and Document Preparation	14
12.1	Basic Document Structure and Styling	14
12.2	Forced Figure Placement and References	14
12.3	Bibliography Citation Syntax	14
12.4	Landscape Orientation in Middle of Document	15
12.5	Correct Command Sequence to Compile LaTeX Documents	15
13	Touch Typing and QWERTY Keyboard Layout	15
13.1	Finger Placement Assignments	15
14	LibreOffice Writer and Document Processing	15
14.1	Default Settings	15
14.2	Clipboard Functions	15
14.3	LibreOffice Usage Scenarios across IT Roles	16
15	LibreOffice Calc and Spreadsheet Formulas	16
15.1	Default Calc Settings and Navigation	16
15.2	Standard Formula Syntaxes	16

1 Computer System Architecture and Fundamentals

1.1 Data vs. Information

- **Data:** Raw, unorganized facts, figures, and symbols supplied as input to a computer system.
- **Information:** Data that has been processed, organized, structured, or contextualized into a meaningful form for human decision-making.

1.2 Characteristics of Computers

- **Speed:** Capable of performing millions of calculations per second.
- **Accuracy:** High precision and error-free operation (errors are typically due to incorrect input data or human instructions).
- **Diligence:** Free from fatigue, lack of concentration, or monotony; operates continuously with consistent accuracy.
- **Versatility:** Capable of performing entirely different types of tasks ranging from word processing to complex data analysis.
- **Power of Remembering:** Ability to store and retrieve huge volumes of data in electronic memory.
- **No I.Q.:** Lacks intrinsic intelligence; executes commands strictly according to programmed instructions.
- **No Feelings:** Devoid of emotion, judgment, or bias.

1.3 Generations of Computers

Generation	Key Technology	Relative Speed	RAM Capacity	External Storage
1st (early 50s)	Vacuum tubes	1	1,000–8,000 bytes	Punch cards
2nd (late 50s)	Transistors	100	4,000–64,000 bytes	Magnetic tape
3rd (60s)	Integrated Circuits	100	32,000–4,000,000 bytes	Magnetic disks
4th (70s+)	Microchips (LSI, VLSI)	1,000,000+	32,000,000+ bytes	Mass storage / Solid state

Table 1: Key Characteristics Across Computer Generations

1.4 Storage Units and Calculations

- **Bit:** Smallest unit of data ((0) or (1)).
- **Byte:** (1 Byte = 8 bits).
- **Kilobyte (KB):** (1 KB = 1024 Bytes = 2^{10} Bytes).
- **Megabyte (MB):** (1 MB = 1024 KB = 2^{20} Bytes).
- **Gigabyte (GB):** (1 GB = 1024 MB = 2^{30} Bytes).
- **Terabyte (TB):** (1 TB = 1024 GB = 2^{40} Bytes).

1.5 System Layers and Operating System Kernel

- **Layered Architecture:**
 1. **Hardware:** Physical components (CPU, RAM, Disks, I/O devices).
 2. **Operating System Kernel:** Core layer interacting directly with hardware.
 3. **System Utilities & Shell:** User interface layer exposing command interpreter.
 4. **Application Software:** User applications (e.g., LibreOffice, Web Browsers).
- **Kernel Definition:** The central component of an operating system that manages system hardware resources (memory, CPU execution time, device control, and file operations) and provides essential services to software applications.

2 GNU/Linux Architecture and Directory Hierarchy

2.1 Filesystem Hierarchy Standard (FHS)

GNU/Linux uses a tree data structure starting at the root directory `/`.

- `/`: Top-most root directory of the Linux directory tree.
- `/dev`: Directory containing special device files representing system hardware (e.g., hard disks, partitions, terminal interfaces).
- `/etc`: Directory containing system-wide configuration files and startup scripts.
- `/boot`: Contains bootloader configuration files, Linux kernel binaries (`vmlinuz`), and initial RAM disk files.
- `/tmp`: Temporary directory where transient files created by applications are stored; files are automatically cleared upon reboot.
- `/var`: Contains variable data files whose content changes dynamically during system operation (e.g., log files, mail queues, database files in `/var/lib`, system backups in `/var/backup`).
- `/home`: Contains user personal home directories (e.g., `/home/username`).
- `/root`: Home directory for the system administrator (`root` user).
- `/media` or `/mnt`: Default mount points for removable storage media (e.g., external USB drives automatically mount under `/media/username/`).

2.2 Path Navigation

- **Absolute Path:** Full explicit file path starting from the root directory `/`.
 - Example: Navigate to `newEN` directory from anywhere:
- **Relative Path:** Navigation path relative to the current working directory, using `.` (current directory) and `..` (parent directory).

```
cd /home/chamath/docs/ugwork/2024/newEN
```

- Example: Moving from `/home/kenneth/documents/backup/` to `/home/kenneth/downloads/tute/intro.pdf`:

```
../../../../downloads/tute/intro.pdf
```

2.3 Symbolic and Hard Links

- **Symbolic Link:** A pointer file that references another target file path.

```
ln -s /var/backup/ug/2023 backup
```

- **Hard Link Removal Behavior:** Removing an original file (`sudath`) linked via hard link to another file (`kamal`) leaves `kamal` intact and fully functional because both hard links point to the same inode on the disk.

3 Linux File Permissions, Ownership, and Security

3.1 Understanding `ls -l` Permission Output

An example output line from `ls -l`:

```
-rwxr-x--- 1 kenneth kenneth 67235 May 31 2019 intro.pdf
drwxr-x--- 1 ruwan    staff   28148 Sep 30 2022 kamal
lrwxr-xr-x 1 ruwan    staff    4096 Sep 30 2022 sudath -> kamal
```

- **1st Character (File Type):**
 - -: Regular file.
 - d: Directory.
 - l: Symbolic link.
- **Permission Triplets (Chars 2–10):** Divided into Owner (u), Group (g), and Others (o).
 - r (Read = 4): View file contents / list directory entries.
 - w (Write = 2): Modify file contents / add or delete files in directory.
 - x (Execute = 1): Run file as an executable script or binary / enter directory.
- **Example breakdown for `-rwxr-x---`:**
 - Owner (`kenneth`): `rw`x (Read, Write, Execute).
 - Group (`kenneth`): `r`-x (Read, Execute).
 - Others: `---` (No permissions).

3.2 Changing Permissions with `chmod`

- **Octal Notation:**
 - `chmod 640 nimal`: Sets permissions to `-rw-r-----` (Owner: Read/Write, Group: Read, Others: None).
 - `chmod 441 kamal`: Sets permissions to `-r--r---x` (Owner: Read, Group: Read, Others: Execute).
- **Symbolic Notation:**
 - `chmod u+x,g-rw sign.sh`: Adds execute permission to owner, removes read and write permissions from group.

3.3 File Ownership Commands

- **Change Group Ownership:**

```
chgrp kamal sign.sh
```

- **Change User Ownership:**

```
chown user filename
```

3.4 Essential Security and Credential Files

- **/etc/passwd:** Stores user account information (username, UID, GID, home directory, login shell).
- **/etc/shadow:** Stores encrypted user passwords and password expiration policies (accessible only by root).
- **/etc/group:** Stores group definitions and member user lists.

3.5 Procedure to Recover Forgotten Root Password

1. Restart the Linux system and enter the GRUB bootloader menu.
2. Select the main kernel option and press **e** to edit kernel boot options.
3. Locate the boot line starting with **linux** and append **init=/bin/bash** or **rw single** at the end.
4. Press **Ctrl+X** or **F10** to boot directly into a single-user root shell.
5. Remount the root filesystem as read-write:

```
mount -o remount,rw /
```

6. Reset the root user password using:

```
passwd
```

7. Reboot the system normally.

4 Disk Management, Partitioning, and File Systems

4.1 Master Boot Record (MBR) Partitioning Types

- **Primary Partition:** A partition that can host an operating system directly. MBR supports a maximum of **four (4)** primary partitions.
- **Extended Partition:** A special primary partition used as a container to hold multiple logical partitions, overcoming the 4-primary partition barrier.
- **Logical Partition:** Secondary partitions created inside an extended partition. In Linux, logical partitions are always numbered starting from index **5** onwards (**sda5**, **sda6**, **sda7**, etc.).

4.2 Disk Device Naming Scheme

- **sda**: Denotes the entire first hard drive disk device.
- **sda1**: Primary partition 1 on disk **sda**.
- **sda5**: First logical partition inside the extended partition.
- **sda7**: Logical partition on disk **sda** (since partition index (≥ 5)).

4.3 Disk Management Commands

- **List partitions and disk geometry:**

```
fdisk -l
```

- **Display available disk space in human-readable units:**

```
df -h
```

- **Check total disk space occupied by current working directory:**

```
du -sh .
```

- **Mount a disk partition to a target folder:**

```
mount /dev/sda4 /home/chamath/docs/ugwork/2025
```

5 Process Management and System Monitoring

5.1 System Monitoring Tools

- **top**: Interactive real-time system performance monitor showing CPU usage, RAM utilization, process hierarchy, load averages, and uptime.
- **ps aux**: Snapshot of currently active running processes.
- **free -h**: Displays total, used, and available system RAM and swap memory.
- **uptime**: Shows system uptime duration, active user sessions, and CPU load averages over 1, 5, and 15 minutes.

5.2 Understanding Load Average Values

Example output line from **top**:

```
top - 23:03:09 up 4 min, 1 user, load average: 0.75, 0.59, 0.25
```

- **0.75**: Average system load over the last **1 minute**.
- **0.59**: Average system load over the last **5 minutes**.
- **0.25**: Average system load over the last **15 minutes**.

5.3 Job Control Key Combinations and Commands

- **CTRL-C:** Interrupts and terminates the active foreground process (SIGINT).
- **CTRL-Z:** Suspends the active foreground process and sends it to background state (SIGTSTP).
- **CTRL-D:** Sends End-of-File signal (logout / close standard input).
- **bg:** Resumes a suspended job in the background.
- **fg:** Brings a background job into the active foreground.

6 Package Management and System Maintenance

6.1 Debian/Ubuntu Package Management (APT)

- **Update repository index:**

```
apt update
```

Updates local package list databases from remote repositories.

- **Meaning of “Hit” in apt update output:** Indicates that the repository index package file has not changed since the last fetch (local cache is up-to-date).
- **Why apt update must be run before searching/installing:** Ensures local repository index cache reflects the newest available package versions and download locations.
- **Search repository for package:**

```
apt search vim
```

- **Install software package:**

```
apt install vim
```

- **Upgrade all software packages:**

```
apt upgrade
```

6.2 RedHat Package Management (RPM)

- **Linux Distributions supporting RPM:**

1. Red Hat Enterprise Linux (RHEL)
2. CentOS
3. Fedora

6.3 Methods of Installing Software on Linux

1. Using official repository package managers (apt, yum, dnf).
2. Installing directly from pre-compiled binary package files (.deb, .rpm).
3. Compiling from raw source code tarballs (./configure && make && make install).
4. Utilizing standalone containerized packages (Snap, Flatpak, AppImage).

7 Networking and System Information Commands

7.1 Networking Commands

- Get system IP address:

```
ip addr
```

(or legacy `ifconfig`).

- Get network interface configuration info:

```
ip a
```

- Display network ports and listening connections:

```
netstat -ntlp
```

7.2 System Hardware Information

- Display Graphics Card Information:

```
lspci | grep -i vga
```

- Display Username of active session:

```
whoami
```

8 BASH Shell Features, Commands, and Pipelines

8.1 Common File Manipulation Commands

- `ls -al`: Detailed long listing including hidden files (files starting with `.`).
- `cat file1.txt file2.txt`: Concatenate and display text content of files.
- `mv file1.txt file2.txt`: Rename `file1.txt` to `file2.txt`.
- `mv file1.txt file2.txt docs/`: Move multiple files into directory `docs/`.
- `head -n12 file.txt`: Display the first 12 lines of `file.txt`.
- `wc file.txt`: Print count of lines, words, and bytes.
- `mkdir docs pics; cd pics`: Create multiple folders and change directory.
- `rm -r docs`: Recursively remove directory `docs` and its content.
- `mkdir -p user/nimal`: Create parent directory `user` and subdirectory `nimal` in a single command execution.
- `touch file1.txt file2.txt file3.txt`: Create multiple empty files or update file timestamps.
- `cat > file.txt`: Create/overwrite file with text typed directly into terminal STDIN.

8.2 BASH Shortcuts and Environment Operators

- `!!`: Re-execute the exact previous command in BASH shell history.
- `!!:p`: Print the previous command without executing it.
- `&&` and `||`:
 - `cmd1 && cmd2`: Logical AND; executes `cmd2` **only if** `cmd1` completes successfully (exit status 0).
 - `cmd1 || cmd2`: Logical OR; executes `cmd2` **only if** `cmd1` fails (non-zero exit status).
- `$?`: Special shell variable holding the return exit status code of the last executed command ((0) denotes success).
- `~/.bashrc`: Shell configuration script executed automatically whenever an interactive shell session starts; used to set aliases and environment variables.
- **Creating Aliases:**

```
alias ls1='ls -l'
alias nstat='netstat -ntlp'
```

Stored persistently in file `~/.bashrc`.

8.3 Advanced Text Processing Pipeline Commands

Given a text file `linux.txt`:

- **Count number of words:**

```
wc -w linux.txt
```

- **List lines ending with a period (.):**

```
grep '\.$' linux.txt
```

- **List lines NOT containing words compute, computing, computation, computer:**

```
grep -v -i -E 'compute|computing|computation|computer' linux.txt
```

- **List distinct words sorted in descending order of frequency:**

```
tr -s ' ' '\n' < linux.txt | tr '[:upper:]' '[:lower:]' | sort |
  uniq -c | sort -nr
```

- **List distinct words sorted by rhyming order:**

```
tr -s ' ' '\n' < linux.txt | rev | sort | rev | uniq
```

9 BASH Scripting and Automation

9.1 Script Positional Parameters

- \$0: Name of the script executable.
- \$1, \$2, \$3: First, second, and third command-line arguments.
- \$*: String representation of all passed arguments.
- \$#: Total count of arguments passed to script.

9.2 Required BASH Script Solutions

9.2.1 Script 1: User Greeting (hello)

```
#!/bin/bash
echo "Hello $USER"
echo "Files in your home directory:"
ls ~
```

9.2.2 Script 2: Word Count Display (whatsin)

```
#!/bin/bash
words=$(wc -w < "$1")
echo "The file(1 has )words words"
```

9.2.3 Script 3: Arithmetic Addition (add)

```
#!/bin/bash
sum=$(( $1 + $2 ))
echo "The sum of $1 and (2 is: )sum"
```

9.2.4 Script 4: File Existence and Permission Checker (files)

```
#!/bin/bash
if [ -e "$1" ]; then
    if [ -x "$1" ]; then
        echo "The file $1 exists and then It is executable"
    else
        echo "The file $1 exists and then It is not executable"
    fi
fi
```

9.2.5 Script 5: Argument Counter and Listing (args)

```
#!/bin/bash
echo "There are $# arguments followed by the text $*"
```

10 Vim Text Editor Syntax and Operations

10.1 Vim Modes

- **Normal / Command Mode:** Default mode for navigation and text editing commands.
- **Insert Mode** (press **i**): For typing text content.
- **Command-Line Mode** (press **:**): For running editor commands, saving, exiting.

10.2 Key Combinations in Normal/Command Mode

- Move cursor to end of line: **\$**
- Move cursor to beginning of line: **0** or **^**
- Move cursor 120 columns left: **120h**
- Move cursor 120 lines down: **120j**
- Move cursor 10 words right: **10w**
- Navigate to top of document: **gg**
- Navigate to bottom of document: **G**
- Cut / Delete current line: **dd**
- Copy / Yank current line: **yy**
- Undo last modification: **u**
- Redo undo modification: **Ctrl+r**

10.3 Window Splitting and File Navigation

- **Vertical split and open new file Banda:**

```
:vsplit Banda
```

- **Horizontal split and open new file Gune:**

```
:split Gune
```

- **Navigate between split views:** Press **Ctrl+w** followed by direction key (**h**, **j**, **k**, **l** or Arrow keys).

10.4 Search and Replace Syntax

- **Search and replace without prompt:**

```
:%s/Chair/Sofa/g
```

10.5 Executing Shell Commands and Key Mapping

- **Insert output of shell command into document:**

```
:r !date  
:r !ls -l
```

- **Map key combination (Ctrl+K) to shell command:**

```
:map <C-k> :!pdflatex file.tex<CR>
```

10.6 Spellchecker Navigation

- **Enable spellcheck:** `:set spell`
- **Navigate forward to next spelling mistake:** `]s`
- **Navigate backward to previous spelling mistake:** `[s`

11 Git Version Control System

11.1 Advantages of Source Code Version Control

1. Tracks complete revision history, allowing restoration of previous file versions.
2. Enables seamless multi-developer collaboration and code branch merging without overwriting work.
3. Acts as an automated remote backup and provides accountability audit logs.

11.2 The Three Areas Defined in Git

1. **Working Directory:** Local checkout directory on disk where files are modified.
2. **Staging Area (Index):** Intermediate buffer storing modified files staged for the next commit.
3. **Local Repository:** Database (`.git/`) containing all committed project snapshots and history.

11.3 Essential Git Commands

- **Initialize repository:** `git init`
- **Check repository status:** `git status` (displays modified, staged, and untracked files).
- **View commit history:** `git log`
- **Compare file differences:** `git diff`
- **Function of Staging Area:** Allows developers to selectively inspect and group specific file modifications before recording them into repository history.
- **Restore file from previous commit:**

```
git checkout <commit_hash> -- filename
```

- **Sharing code without commit history:** Export code via `git archive` or copy working directory files excluding the hidden `.git` folder.

12 LaTeX Technical Writing and Document Preparation

12.1 Basic Document Structure and Styling

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{xcolor}
\usepackage{listings}
\usepackage{natbib}

\begin{document}
\section{Main topic}

Following is a bulleted list of fruits:
\begin{itemize}
  \item Apple
  \item Banana
  \item Cherry
\end{itemize}

\subsection{Sub topic}
% This is a comment not shown in final output
\textbf{This content is in bold format}
\textcolor{red}{color of this text is red}

\end{document}
```

12.2 Forced Figure Placement and References

```
\usepackage{float}

\begin{figure}[H]
  \centering
  \includegraphics[scale=0.8]{pic.png}
  \caption{First Figure}
  \label{fig:pic}
\end{figure}

Figure \ref{fig:pic} is a picture of Randil.
```

12.3 Bibliography Citation Syntax

- In-text Citation Command: `\cite{key}` or `\citep{key}`.
- Bibliography Setup Commands:

```
\bibliography{ex}
\bibliographystyle{ieeetr}
```

12.4 Landscape Orientation in Middle of Document

```
\usepackage{pdflscape}

\begin{landscape}
% Content placed here renders in landscape mode
\end{landscape}
```

12.5 Correct Command Sequence to Compile LaTeX Documents

To build `ex.pdf` containing external listings and bibtex references:

1. `pdflatex ex.tex`
2. `bibtex ex`
3. `pdflatex ex.tex`
4. `pdflatex ex.tex`

13 Touch Typing and QWERTY Keyboard Layout

13.1 Finger Placement Assignments

- **Middle Finger of Right Hand:** Keys 8, I, K, , (comma).
- **Index Finger of Left Hand:** Keys 4, 5, R, T, F, G, V, B.
- **Ring Finger of Left Hand:** Keys 2, W, S, X.

14 LibreOffice Writer and Document Processing

14.1 Default Settings

- **Default File Extension:** `.odt` (OpenDocument Text).
- **Default Document Name:** `untitled1`.
- **Default Page Orientation:** Portrait.
- **Word Processing Component:** LibreOffice Writer.
- **Presentation Component:** LibreOffice Impress.
- **Spreadsheet Component:** LibreOffice Calc.

14.2 Clipboard Functions

- **Cut:** Deletes selected item from its current location and saves it to clipboard storage.
- **Copy:** Duplicates selected item to clipboard without removing it from original location.
- **Paste:** Inserts clipboard contents at active cursor position.

14.3 LibreOffice Usage Scenarios across IT Roles

1. **Computer Science Student:** Writing laboratory reports, assignments, and thesis documentation.
2. **Project Manager:** Preparing project proposals, status reports, and timeline documents.
3. **Quality Assurance Officer:** Authoring test plans, test cases, and software quality reports.
4. **Team Lead:** Drafting technical architecture specifications, design guides, and code review logs.

15 LibreOffice Calc and Spreadsheet Formulas

15.1 Default Calc Settings and Navigation

- **Default File Extension:** .ods.
- **Default Application File Name:** untitled1.
- **Shortcut to reach A1 cell:** Ctrl + Home.

15.2 Standard Formula Syntaxes

Given student mark spreadsheet range E5:H5 for Subject marks, I5 for Average, F5 for Basic Salary:

- **Total Sum:**

```
=SUM(E5:H5)
```

- **Average:**

```
=AVERAGE(E5:H5)
```

- **Conditional Result (Pass if average (≥ 65), else Fail):**

```
=IF(I5>65, "Pass", "Fail")
```

- **Student Rank Calculation:**

```
=RANK(I5, I\ $5:I\ $14)
```

- **Maximum Value:**

```
=MAX(E5:E14)
```

- **Minimum Value:**

```
=MIN(E5:E14)
```


- **Total Number of Students:**

```
=COUNTA(A5:A14)
```

- **Attendance Count:**

```
=COUNT(E5:E14)
```

- **Absence Count:**

```
=COUNTIF(E5:E14, "ab")
```

- **EPF Calculation (8% of Basic Salary):**

```
=F5 * 0.08
```

- **ETF Calculation (3% of Basic Salary):**

```
=F5 * 0.03
```

- **Allowances (10% if Married, else 5%):**

```
=IF(E5="Married", F5*0.10, F5*0.05)
```

- **Gross Salary:**

```
=F5 + I5
```

- **Net Salary (Gross Salary - EPF):**

```
=J5 - G5
```

- **Custom Currency Formatting Code:**

```
"Rs. "#,##0.00
```